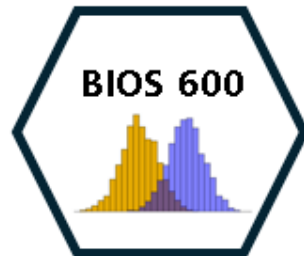


Lecture 17: Introduction to linear regression

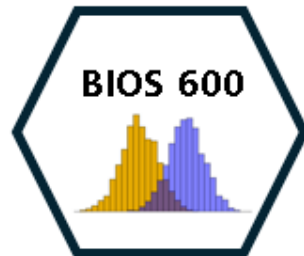
BIOS 600 - Fall 2025

UNC - Chapel Hill



Announcements

- ▶ Let's go over the project instructions!
- ▶ Project proposals due Tuesday, Nov 11 at 11:59pm
 - ▶ Only one person should turn in the proposal per group (On Canvas). Other group members should write in the Text Entry box: [Group Member] turned in the proposal for our group.
- ▶ Video upload instructions will be posted tomorrow (Friday).
- ▶ No HW until next week.



Uploading an excel file

- ▶ You can upload .xlsx files using the **readxl** package.

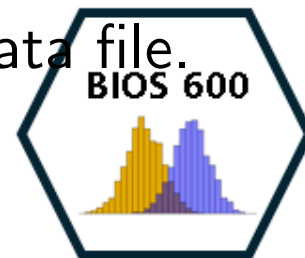
```
# Install (only once)
install.packages("readxl")

# Load each time you start R
library(readxl)

# read in data

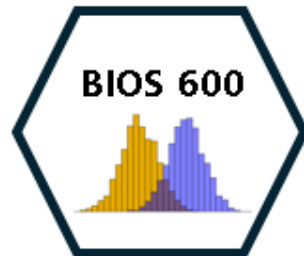
# Example: import a dataset called "biostats_data.xlsx"
data <- read_excel("data/biostats_data.xlsx")
```

- ▶ Let me or the TAs know if you need help reading in a data file.

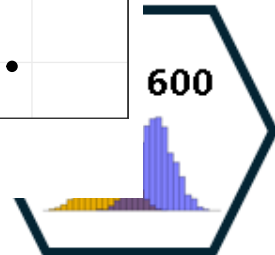
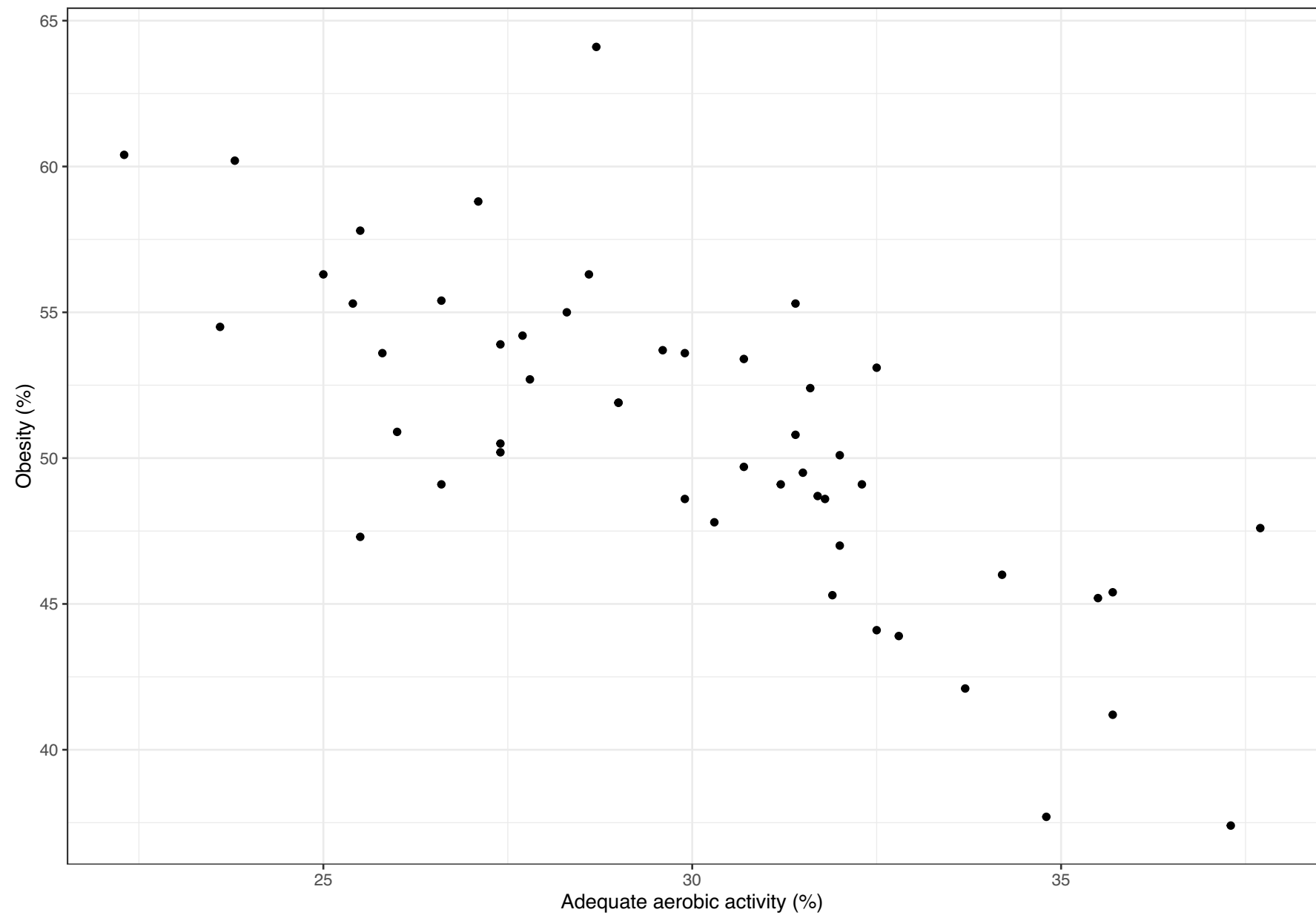


Reading

- ▶ P&G Chapter 18
- ▶ OI: Chapter 8

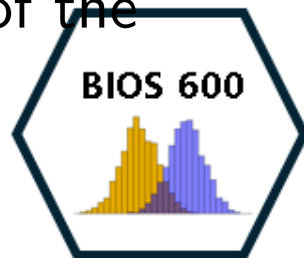


Back to Lab 2...



Correlation and regression

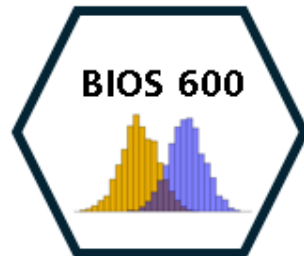
- ▶ Correlation measures the direction and strength of linear relationships
- ▶ Regression can be used to further quantify these relationships
- ▶ A **regression** line summarizes the relationship between explanatory or predictor variables (e.g., Exercise % or X) and response or outcome variables (e.g., Obesity % or Y)
- ▶ The fitted regression line can be used to **predict the outcome** for a given set of predictor values. Our predictions do have error, called **residuals** (vertical distance between observation and the regression line).
- ▶ The **least-squares regression line** minimizes the sum of the squared errors.



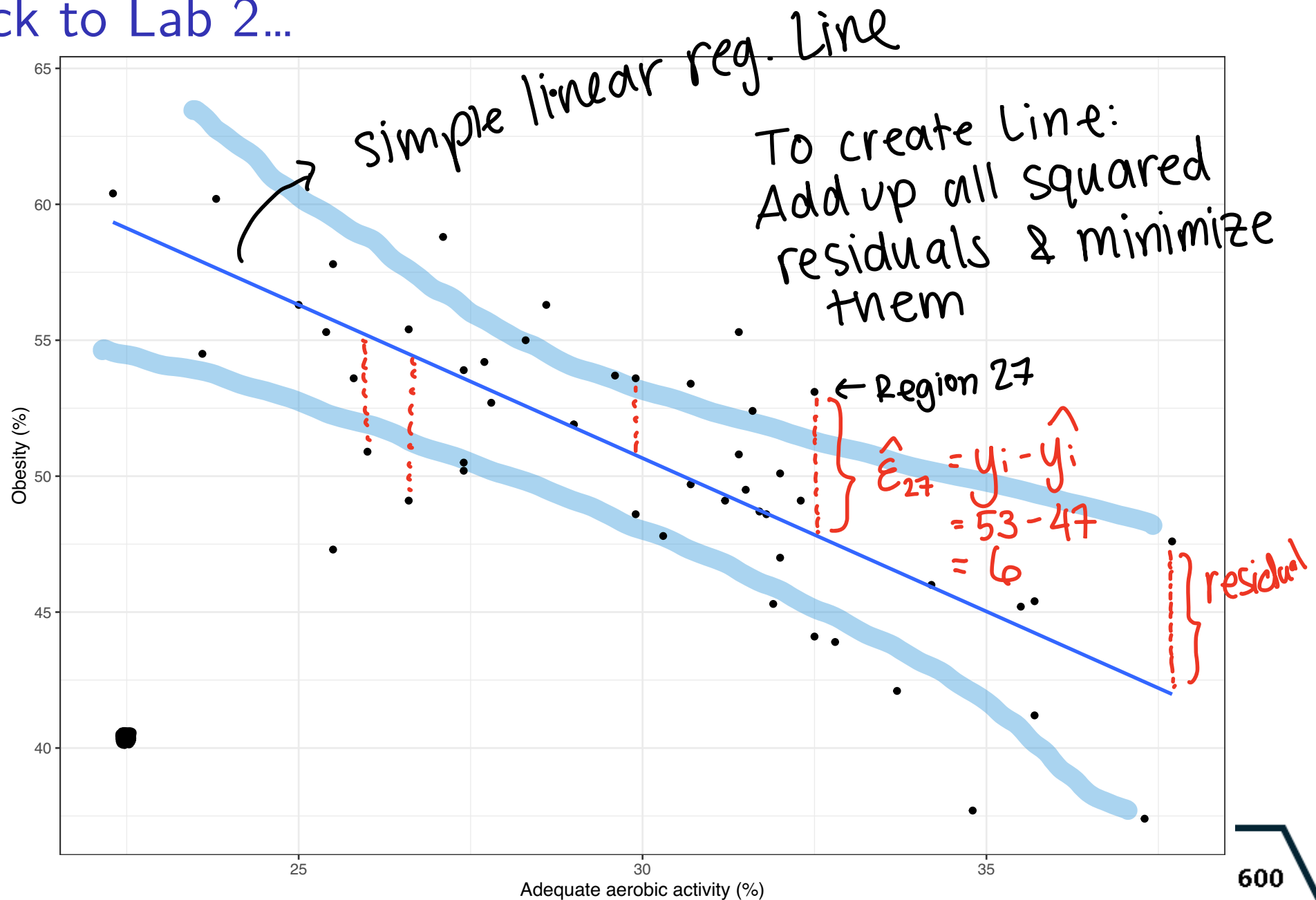
Regression

simple linear reg: only 1 predictor
multiple " " : more than 1 predictor

- ▶ In regression, we use one variable (or more) to try to predict values of another.
- ▶ In simple linear regression, one variable Y is the **response** or **outcome** or **dependent variable** and the other X is the **predictor** or **explanatory** variable or **independent** variable.
- ▶ This distinction is critical. The regression of Y on X is **not equal** to the regression of X on Y .
- ▶ The regression of Y on X can be used to predict Y based on fixed values of X .



Back to Lab 2...



- ▶ We see the regression line relating exercise and obesity %
- ▶ Points represent actual data values; note that the line is not a perfect fit

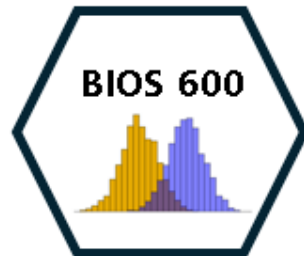
Code for previous slide

```
cdc |>
  ggplot(aes(x = Obesity, y = Exercise)) +
  geom_point() +
  theme_bw() +
  labs(x = "Adequate aerobic activity (%)",
       y = "Obesity (%)") +
  geom_smooth(method = "lm", se = F)
```

↑
linear
model

↑
std.
error

(confidence
band)



The linear model

1st person: $y_1 = \beta_0 + \beta_1 x_1 + \varepsilon_1$

2nd person: $y_2 = \beta_0 + \beta_1 x_2 + \varepsilon_2$

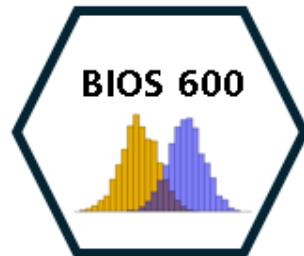
3rd person: $y_3 = \beta_0 + \beta_1 x_3 + \varepsilon_3$

→ epsilon

The model is given by $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$, where

- ▶ y_i is the outcome (dependent variable) of interest
- ▶ β_0 is the intercept parameter
- ▶ β_1 is the slope parameter
- ▶ x_i is the predictor variable
- ▶ ϵ_i is the error (like β_0 and β_1 , it is not observed)

The i is an index of observations. So, each observation of the outcome is related to an observation of that individual's value of the predictor.



The least squares model

Fitted line equation:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

Residual:

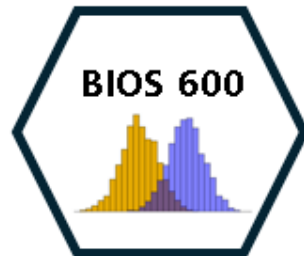
$$\hat{\epsilon}_i = y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i) = y_i - \hat{y}_i$$

observed y-value
= predicted y-value

Least squares selects $\hat{\beta}_0$ and $\hat{\beta}_1$ such that the **error sum of squares**

$$\sum_{i=1}^n \hat{\epsilon}_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

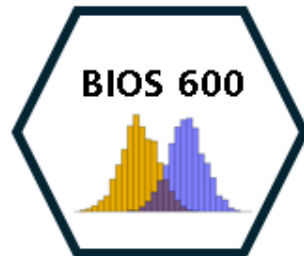
is minimized.



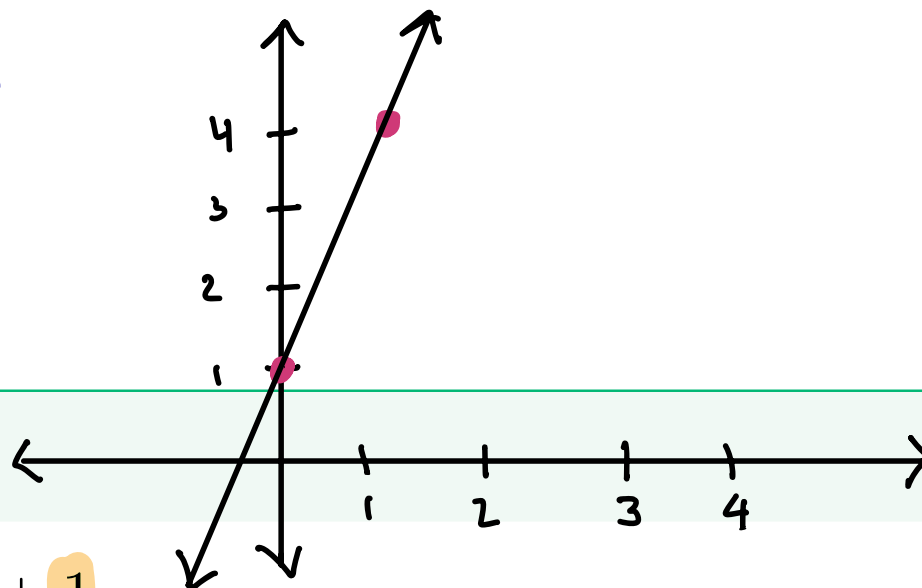
Review slope-intercept form of a line

Recall the slope-intercept form of a line, $y = mx + b$.

- ▶ b is the y-intercept, or where the line crosses the y-axis. It is the predicted value of y when $x = 0$
- ▶ m is the slope, which tells us the **predicted increase** in y when x **changes by 1 unit**



Review: formula of a line



💡 Review

Consider the line $y = 3x + 1$.

- ▶ What value does y take when $x = 0$?
- ▶ As x increases by 1 unit, how much do we expect y to increase by?
- ▶ In statistics, we often represent the **slope** with β_1 and the **intercept** with β_0 , and these values are usually not known but must be estimated.

Model output γ $\times$

```
fit <- lm(Obesity ~ Exercise, data = cdc)
summary(fit)
```

Call:

```
lm(formula = Obesity ~ Exercise, data = cdc)
```

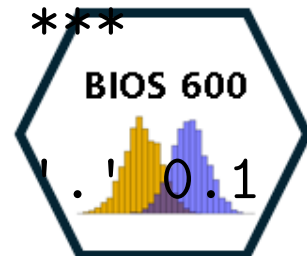
Residuals:

Min	1Q	Median	3Q	Max
-6.118	-1.150	-0.065	1.385	6.229

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	54.79087	3.24772	16.871	< 2e-16 ***
Exercise	-0.48991	0.06365	-7.697	6.34e-10 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1



Residual standard error: 0.455 on 48 degrees of freedom

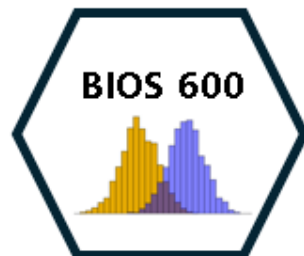
A neater way

```
library(tidymodels)
library(kableExtra)

tidy(fit) |>
  kable()
```

term	estimate	std.error	statistic	p.value
(Intercept)	54.7908720	3.2477165	16.870583	0
Exercise	-0.4899056	0.0636478	-7.697129	0

► Can we do better?



Even better

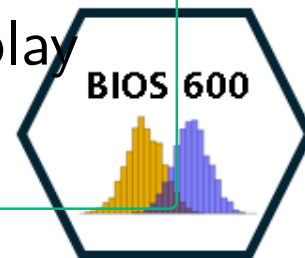
```
tidy(fit) |>  
  kable(digits = 3) # round to 3 digits
```

	term	estimate	std.error	statistic	p.value
$\hat{\beta}_0$	(Intercept)	54.791	3.248	16.871	0
$\hat{\beta}_1$	Exercise	-0.490	0.064	-7.697	0

- Note: p-values are not actually 0. They are just quite small so they get rounded to zero in the table.

💡 Tip

- If you include a regression model in your project, display the output with `tidy()` and `kable()` like this!



Intercept and slope

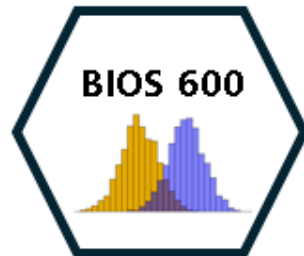
- ▶ The **intercept** is the estimated mean outcome when the predictor is zero (does this make sense in context?)
- ▶ The **slope** is the expected change in the outcome corresponding to a one-unit change in the predictor

β_1 : **Slope Interpretation:** For each percentage increase in Exercise, we expect the Obesity **percentage** to decrease by .48., *on average.*



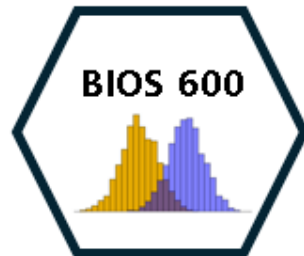
Tip

For all interpretations, remember to **include units!**



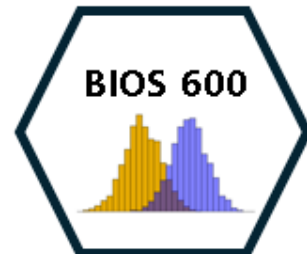
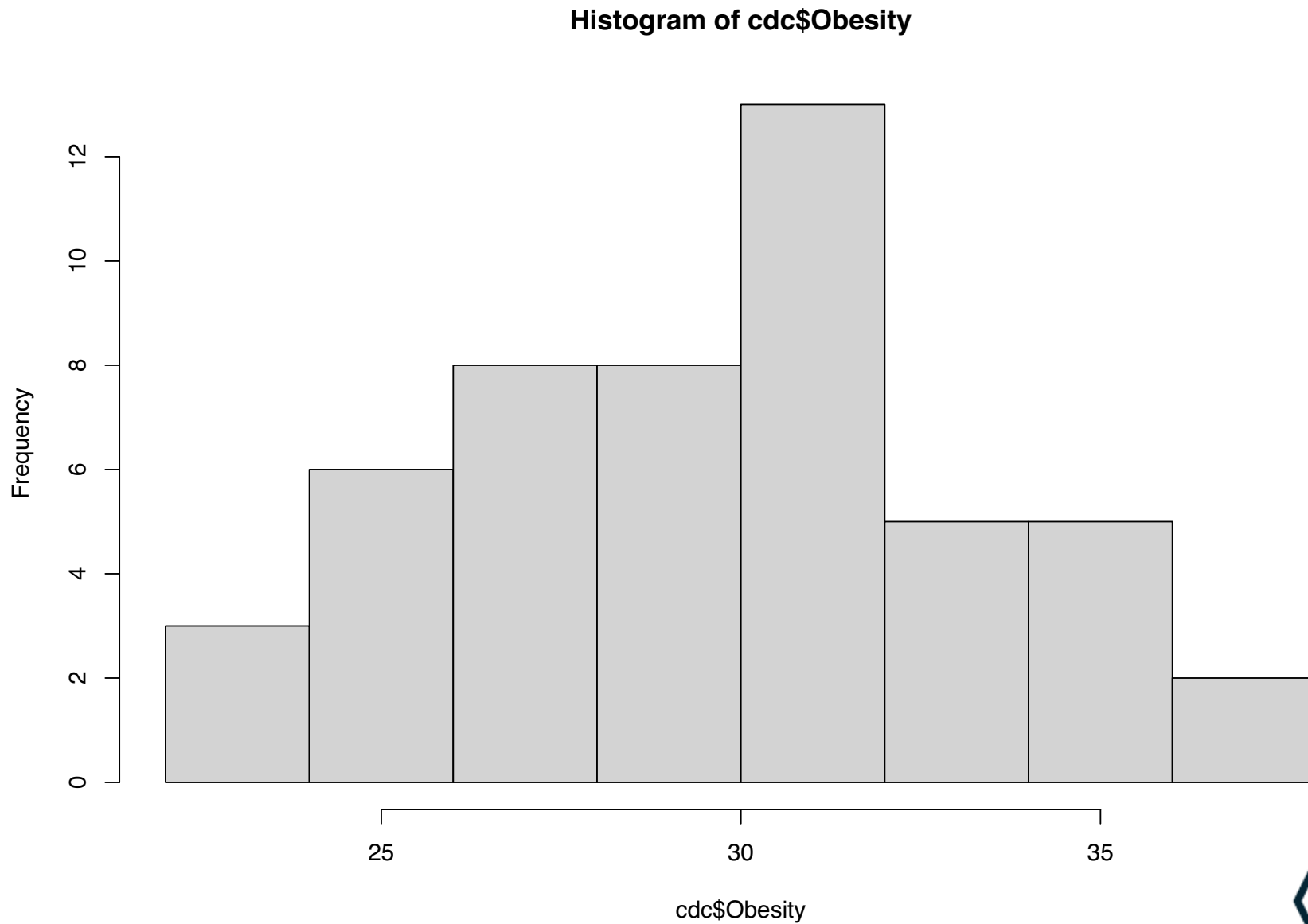
Intercept interpretation

- ▶ **The technical intercept interpretation:** When zero percent of the region gets adequate Exercise, we would expect the percent of the region that is obese to be 54.79%.
- ▶ Does it make sense to interpret the intercept here?
- ▶ You should only interpret the intercept in a regression model if:
 - ▶ the predictor can **feasibly** take values equal to or near zero, or
 - ▶ there are values near zero in the observed data



Let's check

hist (cdc \$ Obesity)

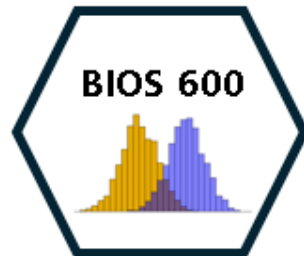


Let's check

```
min(cdc$Obesity)
```

```
[1] 22.3
```

- ▶ Since the predictor (`Obesity`) does not feasibly take on values equal to zero (and we also checked that the lowest value in the dataset is 22.3%), we should not interpret the intercept.

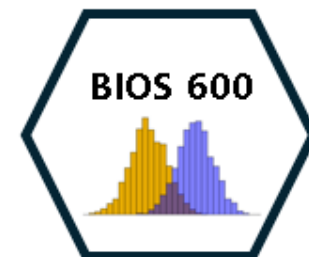


Model assumptions

Assumptions of the model $y_i = \beta_0 + \beta_1 x_1 + \epsilon_i$ include:

- ▶ The outcomes y_i are independent. This is violated if our study contains repeated outcome measures on an individual, if siblings are enrolled, etc.
- ▶ A linear relationship holds (though we can relax this to some extent)
- ▶ For a specified value of x , which is measured without error, $y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$. Note that this implies $\epsilon_i \sim N(0, \sigma^2)$.
- ▶ The variance σ^2 is constant across all values of x ; this is called **homogeneity of variance**

$$y_i \sim N(\mu = \beta_0 + \beta_1 x_i, \sigma^2)$$



Hypothesis testing

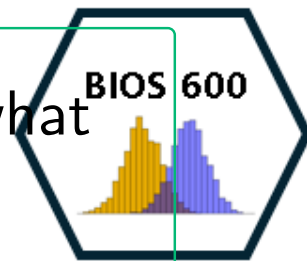
- ▶ The primary hypothesis test of interest is usually $H_0 : \beta_1 = 0$ (There is **no association** between exercise and obesity percentage) versus $H_1 : \beta_1 \neq 0$ (there is an association between exercise and obesity percentage.)
- ▶ Under the null hypothesis, we can use a t-test to test this.
- ▶ R automatically tests this with the `lm()` function, and provides the t-statistic and p-value in the output.

	term	estimate	std.error	statistic	p.value
	(Intercept)	54.791	3.248	16.871	0
*	Exercise	-0.490	0.064	-7.697	0



Given our t-statistic of -7.7 and p-value of <0.001 , what might we conclude in our data?

Reject H_0



Some additional output

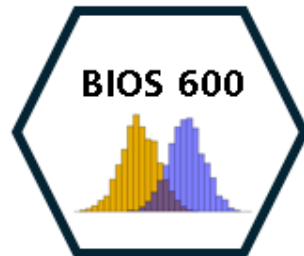
We have statistically significant evidence that there is an association btwn Exercise % and Obesity %, at $\alpha = .05$ level.

```
summary(fit)$r.squared
```

```
[1] 0.55243
```

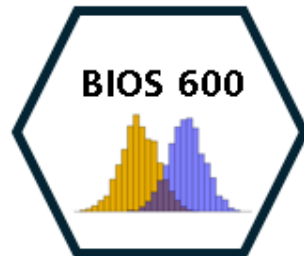
- ▶ The R – *squared* value indicates that exercise explains 55% of the variance of obesity.
- ▶ R^2 tells us about how much of the variability in obesity is explained by exercise; as it gets close to 1, the points get tighter around the regression line.

R^2 ranges from 0 → 1



Predicted values

- ▶ The regression equation can be used to get predicted means at any value of x .
- ▶ $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$, where we call $\hat{y}$ the predicted value of y at a given value of x (what happened to the error term?).



Predicted values

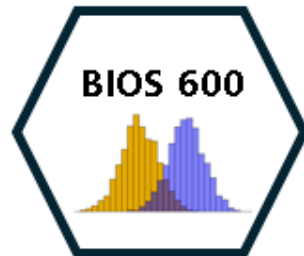
- For the CDC data that predicts a state's obesity percentage by its adequate exercise percentage, we have

$$\hat{y}_i = 54.79 - 0.49x_i$$

So, the predicted mean obesity percentage for a state where 50% of residents get adequate exercise is

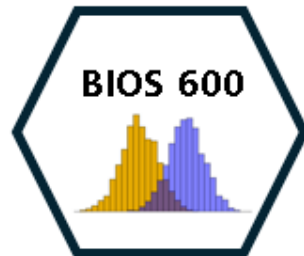
$$\hat{y}_i = 54.79 - 0.49(50)$$

We would predict an obesity percentage of approximately 30%.
Note the importance of getting the units correct.



Recap

- ▶ What is a linear regression model? How do we fit & visualize the model in R?
- ▶ Slope, intercept and interpretations
- ▶ Understand when it's appropriate to interpret the intercept
- ▶ Interpret R^2 from regression output



Next class

- ▶ We will learn how to check the regression model assumptions in R
- ▶ Multiple regression

